INCH-POUND

MIL-DTL-7601C
17 October 2011
SUPERSEDING
MIL-G-7601B
15 November 1982

DETAIL SPECIFICATION

GAGE, EMERGENCY OXYGEN, TYPE L-2

Reactivated after 17 October 2011 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of high-pressure, emergency oxygen gage, range 0 to 2,500 pounds-force per square inch gauge (psig), designated type L-2.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to DLA Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.daps.dla.mil/.

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FEDERAL STANDARD

FED-STD-595/31136	- Red, Flat or Lusterless

COMMERCIAL ITEM DESCRIPTION

A-A-59503	- Nitrogen, Technical

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-27210	- Oxygen, Aviator's Breathing, Liquid and Gas

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-889	- Dissimilar Metals
MIL-STD-31000	- Technical Data Packages

(Copies of these documents are available online at https://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B545	- Standard Specification for Electrodeposited Coatings of Tin

(Copies of this document are available online at http://www.astm.org/ or from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE AS71051	- Pipe Threads, Taper, Aeronautical National Form, Symbol ANPT - Design and Inspection Standard
SAE AS8879	- Screw Threads - UNJ Profile, Inch Controlled Radius Root with Increased Minor Diameter
SAE AS5201	- Fitting End, External Taper Pipe Thread, Design Standard

(Copies of these documents are available online at http://www.sae.org or from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article inspection. When specified (see 6.2), sample gages shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials.

3.2.1 Nonferrous materials. Unless otherwise specified, nonferrous materials shall be used for all parts of the gage except where ferrous materials are essential. Ferrous materials, when used, shall not come in direct contact with oxygen.

3.2.1.1 Dissimilar materials. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

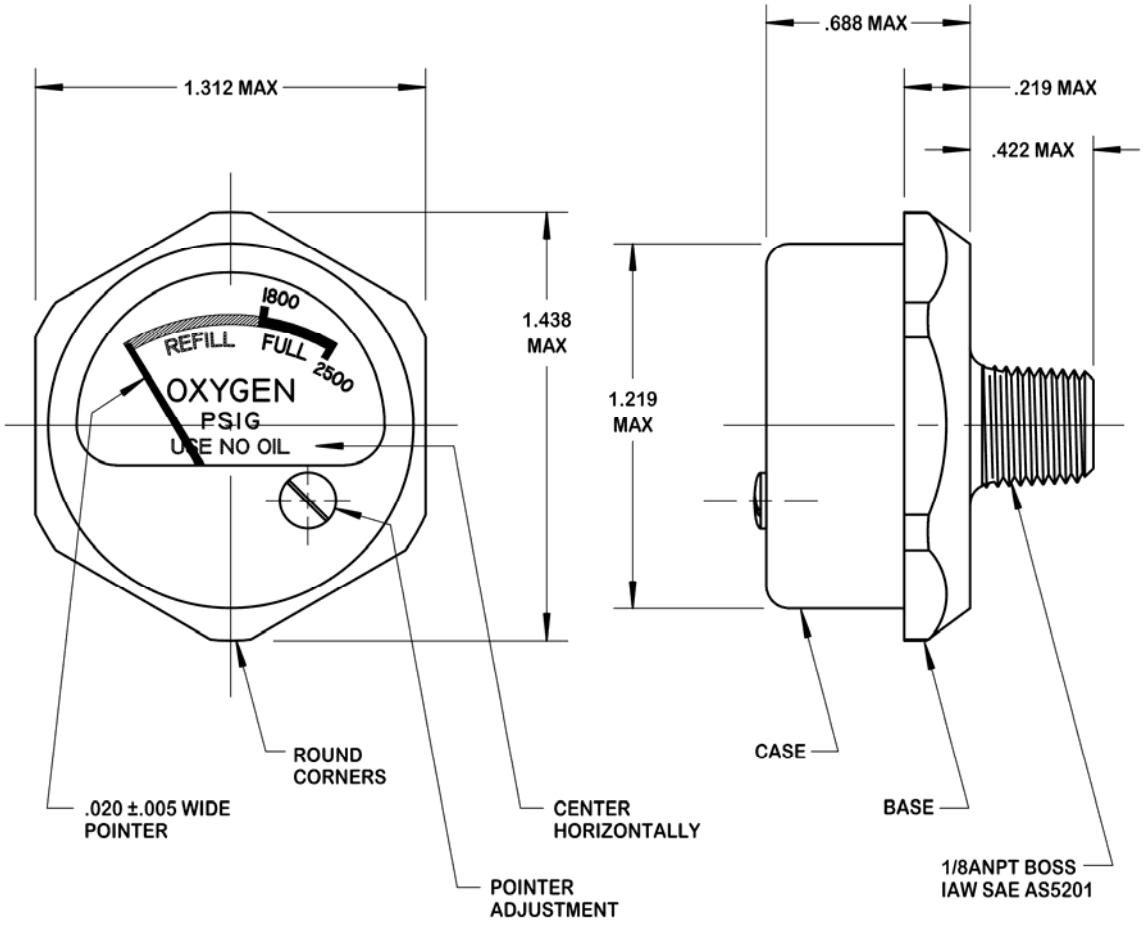
3.2.2 Nonmetallic materials. Any nonmetallic material that is easily deteriorated or otherwise easily affected adversely by continued service use with oxygen shall not be used.

3.2.3 Protective treatment. When materials are used in the construction of the gage that are subject to corrosion in oxygen, salt air, or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coatings and finishes that will crack, chip, or scale with normal service life, age, or extremes of atmospheric conditions shall not be used.

3.2.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design. The gage shall be designed to conform to figure 1.

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NOTES:
1. All dimensions are shown in inches.
2. Unless otherwise specified, all tolerances shall be ±0.016.
3. Dial text and numerical markings shall be not less than 0.062 in height.
4. 1/8ANPT fitting shall be tinned in accordance with ASTM B545.
5. The left-most end of the dial range (below 1800), the "REFILL", and the "USE NO OIL" markings shall be colored a flat, lusterless red to conform with FED-STD-595/31136.
6. All other dial markings and the pointer shall be colored flat black.
7. The left edge of the pointer shall be considered the indicating edge.
8. The dial face shall have a white background.

FIGURE 1. Gage, emergency oxygen, type L-2.

3.4 Construction. The gage shall be so constructed that no parts will work loose in service. The mechanism shall be built to withstand without failure, the strains, jars, vibration, and other conditions incident to shipping, storage, installation, and service. Pivots, bearings, and gears, shall neither bind nor shake, and shall be as near frictionless as practicable. The gages shall be so constructed that minor repairs such as replacement of cover glasses and pointers can be easily made by the personnel of the operating units without the use of special tools.

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3.4.1 Hair springs. If hair springs are used in the construction of the gage, they shall be made of beryllium copper or phosphor bronze.

3.4.2 Bourdon tube. The bourdon tube shall be constructed of beryllium copper.

3.4.3 Viewing window. The viewing window shall be constructed of transparent plastic materials, clear, flat, and free from flaws which will affect the readability of the gage. The plastic material shall not craze and shall remain transparent when subjected to the temperatures specified herein.

3.4.4 Case.

3.4.4.1 Base. The base of the case shall be fabricated from brass or other suitable metal.

3.4.4.2 Cover. The cover of the case shall be fabricated from steel or other materials of suitable strength to withstand the normal service conditions. Materials such as soft aluminum which are subject to damage under service conditions shall not be used.

3.4.5 Protective cap. The external thread fitting conforming to SAE AS5201 shall be fitted with a suitable cap to protect the threads and to prevent dust and foreign matter from entering the gage during shipping and storage.

3.4.6 Pointer. The pointer shall be of sufficient length that the top end will extend not less than half of the distance into the dial arc marking. The pointer shall be dull black in color. The pointer shall be 0.020 ±0.005 inch wide at the point where it crosses the lower scale arc. The pointer shall be as light as practicable and sufficiently rigid to prevent oscillation under vibration.

3.4.6.1 Pointer adjustments. A mechanical high limit stop, if used, shall be applied to the mechanism and not to the pointer. The pointer shall be firmly attached to the mechanism, yet shall be readily adjustable.

3.4.7 Dial. The dial shall be securely fastened upon the frame assembly to insure that it will not become loose or turn when the gage is subjected to vibration.

3.4.8 Dial to viewing window distance. The distance between the dial and the inside of the viewing window shall be the minimum practicable, but shall be sufficient to insure freedom of movement of the pointer under all conditions of gage operation.

3.4.9 Mechanism adjustment. The mechanism shall be provided with a means to adjust the gage indication, as necessary, during the service life of the gage. The means of adjustment shall be reasonably simple to manipulate, and shall be of such a nature that adjustment can be accomplished without the use of special tools.

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3.4.10 External construction. The indicating and pressure elements shall comprise a sub-unit distinct from the case. The gage external construction, to include the case, the base, and the threaded fitting, shall be sufficiently rugged to insure that small distortions of the case will not have undue effect on the internal sub-unit, thereby affecting the accuracy of the gage indication.

3.5 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part number shall be governed by the technical data requirements of MIL-STD-31000.

3.6 Weight. The weight of the gage shall not exceed 0.16 pound.

3.7 Threads.

3.7.1 Screw. Machine screw threads shall be in accordance with SAE AS8879.

3.7.2 Pipe. ANPT pipe threads shall be in accordance with SAE AS71051. Threads shall be tinned in accordance with ASTM B545.

3.7.2.1 Pipe thread lubrication. Unless otherwise specified (see 6.2), the pipe thread shall be free of lubrication.

3.8 Filter. The inlet shall be provided with a sintered bronze porous plug to prevent foreign materials from entering the gage.

3.9 Performance.

3.9.1 Leakage.

3.9.1.1 Bourdon tube. The bourdon tube and base, upon completion of the assembly, shall be capable of withstanding a gas pressure of 2,000 psig without leakage while immersed in a water bath.

3.9.1.2 Case. The case shall be mechanically tight in all joints.

3.9.2 Scale error. A master gage of known accuracy shall indicate at least 1,800 psig, but not more than 1,900 psig, when the gage being tested indicates 1,800 psig.

3.9.3 Friction. The pointer shall not move more than the width of the black indication marker (see figure 1) when a pressure of 1,800 psig is applied and the gage is tapped lightly. The pointer shall move smoothly throughout the complete deflection range without vibration.

3.9.4 Drift. The pointer of the gage shall not move out of the width of the black indication marker (see figure 1) at 1,800 pounds after being subjected to 1,800 psig pressure for 1 hour. The gage shall be lightly tapped during this test.

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3.9.5 Seasoning. The gage shall be capable of satisfactory operation after being subjected to a pressure of 1,800 psig for not less than 300 cycles.

3.9.6 Temperature.

3.9.6.1 Low temperature. The gage shall demonstrate proper function at a temperature of -65 °F (-54 °C).

3.9.6.2 High temperature. The gage shall demonstrate proper function at a temperature of 160 °F (71 °C).

3.9.7 Overpressure. The gage shall not leak or burst while being subjected to a hydrostatic pressure of 3,500 psig for not less than 5 minutes.

3.9.8 Vibration. The gage shall function satisfactorily after being vibrated at any desired frequency between 500 and 2,500 cycles per minute (cpm) with a double-amplitude of not less than 0.018 inch, nor more than 0.020 inch, for a period of not less than 3 hours.

3.10 Markings.

3.10.1 Dial markings. The dial shall have a white background with dull black markings, unless otherwise specified (see 6.2). It shall be marked with an arc 0.031 inch wide showing the pressure range of the gage. A black indication marker, consisting of a segment of a width corresponding to a 100-psig segment of the scale arc plus the width of the pointer at the scale arc, shall be placed at the point corresponding to the 1,800 psig pressure indication on the arc. This marker shall be set normal to the upper scale arc surface. True 1,800-psig pressure shall be indicated on the gage by the left-most edge of the pointer lining up with the left-most edge of the 1,800-psig indicator marking. A marking of "1800" (less quotes) shall be plainly marked in proximity to the 1,800-psig pressure indication marker on the dial.

3.10.1.1 Up-pressure side. The remainder of the indicator scale on the up-pressure side shall be marked in black and continue to the 2,500-pound pressure limit. A marking of "2500" (less quotes) shall be plainly marked in proximity to the 2,500-psig pressure indication marker on the up-pressure limit end of the dial.

3.10.1.2 Indicator scale. Starting at the left edge of the 1,800-psig pressure indication marking, the indicator scale shall be continued to the zero-pressure limit in a lusterless red color, in accordance with shade number 31136 of FED-STD-595. The words "REFILL" (less quotes) shall be in red, and "FULL" in dull black and shall be located along the arc as shown in figure 1. The word "OXYGEN" shall be marked on the dial in dull black. The "FULL" and "REFILL" markings may be located on the opposite side of the arc instead of being located as shown on figure 1.

3.10.1.3 Additional markings. The following marking in red letters shall be placed on the dial as shown on figure 1: "USE NO OIL" (less quotes).

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3.10.1.4 Durability. All markings shall be sufficiently durable to withstand use encountered in normal service life.

3.11 Identification of product. Each gage shall be marked in accordance with MIL-STD-130, with the following information, using letters not less than 0.062 inch high:

GAGE, EMERGENCY OXYGEN
TYPE L-2 / MIL-DTL-7601
MFR: <Manufacturer CAGE number>

3.11.1 Use of MIL designation. MIL designation shall not be applied to a product, except for first article test samples, nor referred to in correspondence, until notice of approval has been received from the activity responsible for first article tests.

3.12 Workmanship. Workmanship shall be in accordance with high-grade aircraft instrument manufacturing practices. The gage shall be thoroughly cleaned of all oil and foreign matter at the proper stage of assembly.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Test conditions.

4.2.1 Atmospheric and temperature. Unless otherwise specified, all tests required by this specification shall be made at an atmospheric pressure of approximately 29.92 inches of mercury and an ambient temperature of approximately 68 °F (20 °C). When tests are made with ambient pressure or temperature different from the above values, proper allowance shall be made for the difference from the specified conditions.

4.2.2 Gas. The high-pressure gas used in testing the gage shall be either oxygen conforming to MIL-PRF-27210, oil-free nitrogen conforming to A-A-59503, or oil-free air equivalent in dryness to A-A-59503 nitrogen.

4.2.3 Tapping. Unless otherwise specified, the gage shall be tested in normal vertical operating position and shall be lightly tapped before a test reading is taken.

4.2.4 Vibration. Vibration shall be at any desired frequency between 500 and 2,500 cpm and at a double-amplitude of not less than 0.018 inch, nor more than 0.020 inch.

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4.2.5 Tolerance. To provide for changes in the gage mechanism due to expansion and contraction of parts when testing for scale errors under conditions of high and low temperatures specified herein, the following deviation from scale error test will be allowed. During the rechecking for the high and low temperature scale tests, a deviation will be allowed whereby the master gage shall indicate at least 1,800 psig, but not more than 1,950 psig.

4.3 First article tests.

4.3.1 First article test. First article test samples shall consist of four gages to be tested in accordance with tests specified herein.

4.3.2 Tests. The first article tests of gages shall consist of all the tests specified under 4.4.

4.4 Conformance tests. Conformance tests shall consist of individual tests (see 4.4.1) and sampling tests (see 4.4.2).

4.4.1 Individual tests. Each gage shall be subjected to the following tests.

4.4.1.1 Examination of product. Unless otherwise specified by the procuring activity (see 6.2), each gage shall be examined to determine conformance with this specification, with respect to materials, fit, workmanship, and markings.

4.4.1.2 Leakage tests.

4.4.1.2.1 Bourdon tube and base. After assembly of the Bourdon tube and base and before attachment of the dial and pointer, each sub-assembly shall be tested at a gas pressure of 2,000 psig to insure absolute freedom from leaks as evidenced by no bubbles in a soap and water solution applied over the surface of the Bourdon tube assembly.

4.4.1.3 Scale error. The gage being tested and a master test gage of known accuracy shall be connected in parallel to a pressure source, and pressure shall be applied until the left-most edge of the pointer on the sample gage indicates 1,800 psig. At this point, the master gage shall indicate not less than 1,800 psig, nor more than 1,900 psig. The pressure on the master gage shall be increased to an indication of 2,000 psig. The pressure shall then be bled down until the pointer on the gage under test indicates 1,800 psig. At this point, the master gage shall indicate not less than 1,800 psig, nor more than 1,900 psig.

4.4.1.4 Friction. The gage shall be tested for friction at 1,800 psig in the following manner: Pressure shall be increased on the gage until the pointer indicates a pressure of 1,800 psig. At this point, the gage shall be tapped and the pointer shall not move out of the width of the 1,800-psig graduation on the gage scale (see figure 1). The pointer shall move smoothly while the pressure is varied uniformly without vibration of the gage. This test may be combined with the scale error test.

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4.4.2 Sampling tests. Samples shall be selected as specified in sampling plan A (see 4.4.2.1) or sampling plan B (see 4.4.2.2) and shall be subjected to the tests described in the applicable paragraph. A lot shall consist of all gages manufactured under essentially the same conditions at substantially the same time, and from identical materials.

4.4.2.1 Sampling plan A. Five gages shall be selected at random from each lot and subjected to the following tests.

4.4.2.1.1 Vibration and low temperature. The gage shall be mounted on a suitably designed vibration stand and subjected to a temperature of -65 °F (-54 °C). While at this temperature, the gage shall be subjected to vibration for a period of 3 hours, duplicating as nearly as practicable the vibratory conditions found in an actual installation. The vibration shall be as specified herein and at a frequency of 500 to 2,500 cpm. During this test, a minimum gas pressure of 1,800 psig shall be maintained on the gage.

4.4.2.1.1.1 Scale errors. The gage shall be tested for scale errors as specified in 4.4.1.3, after the 3-hour period and after vibration has been stopped but the gage is still at the specified temperature. The tolerance specified in 4.2.5 shall apply during this test.

4.4.2.1.1.2 Leaks. After the preceding scale error and vibration at low temperature test, the gage shall be tested for leaks as specified herein.

4.4.2.1.2 High temperature. The gage shall be subjected to a temperature of 180 °F (82 °C) for a period of 3 hours, after which it shall again be tested for scale error and leakage as specified herein. The tolerance specified in 4.2.5 shall apply during this test.

4.4.2.1.3 Drift. The gage shall be subjected to and shall indicate a pressure of 1,800 psig. The pressure shall be maintained for 1 hour. After the hour has passed, the 1,800-psig indication on the gage shall not have increased by more than the width of the 1,800-psig graduation.

4.4.2.1.4 High pressure. The gage being tested and a master gage of known accuracy shall be connected in parallel and subjected to a source of pressure. Pressure shall then be applied until the pointer of the sample gage indicates 1,800 psig. At this point, the master gage shall indicate not less than 1,800 psig, nor more than 1,900 psig. The pressure on the master gage shall then be increased to indicate 2,500 psig. The pressure shall then be bled down until the pointer on the gage being tested indicates 1,800 psig. At this point, the indication on the master gage shall indicate not less than 1,800 psig, nor more than 1,950 psig.

4.4.2.2 Sampling plan B. One gage shall be selected at random from each lot and shall be subjected to the following tests.

4.4.2.2.1 Overpressure. The gage shall be hydrostatically overpressurized to 3,500 psig for a period of 5 minutes without bursting. The gage shall then be thoroughly dried and again subjected to the leakage test (see 4.4.1.2). Gages subjected to this test shall not be submitted as part of the contract or order.

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4.4.2.2.2 Seasoning. The gage shall be subjected to 300 pressurization cycles extending from 0 psig to 1,800 psig and back to 0 psig at a rate of approximately 1 cpm. Twenty-four hours after the completion of this test, the gage shall satisfactorily pass the scale error test (see 4.4.1.3).

4.4.3 Rejection. Any gage failing to meet the requirements of the individual test shall be rejected and returned at the contractor's expense. Any sample that fails to meet the requirements of the sampling tests (see 4.4.2) shall result in the lot represented being rejected and returned at the contractor's expense.

4.4.3.1 Quantity representative failure. When tests are specified on a quantity of gages that are selected as representative of a certain lot, and one or more of this number fails to meet the requirements, additional gages of the lot represented shall be tested immediately to determine the cause of the failure. Individual performance tests shall not be interrupted unless the defect is of such a nature that it will seriously affect the performance or safe use of the gage.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The type L-2 emergency oxygen gage covered by this specification is used with the type H-2 cylinder assembly and is intended principally to show whether or not the cylinder is fully charged and is also used to read accurately at the 1,800-psig mark.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. First article testing, if required (see 3.1).
c. Pipe thread lubrication, if required (see 3.7.2.1).
d. Variations in dial marking coloration (see 3.10.1).
e. Product examination requirements (see 4.4.1.1).
f. Packaging requirements (see 5.1).

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6.3 Subject term (key word) listing.

ANPT
Antiseize
Beryllium copper
Bourdon
Dial
H-2 cylinder
Hydrostatic
Master gauge
O2 system
Oil free
Overpressure
Phosphor bronze
Polytetrafluorethylene
Pressure
Psig
Teflon

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy - AS
Air Force - 99
DLA - GS

Preparing Activity:
DLA - GS1

(Project 6685-2011-012)

Review Activity:
Air Force - 71

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